Simul Ahmed

Embedded Systems & PCB Design Engineer

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PROFILE

Embedded systems and PCB design engineer with full-cycle hardware experience, from schematic and layout in Altium Designer to fabrication outputs and production support. Core R&D contributor to a commercial power-control product with 2,000+ units deployed, designer of a four-board industrial edge system for a contract client, and member of the team behind a healthcare IoT product. Altium-certified, with firmware-aware layout experience on STM32, ESP32 and ATmega. Final-year B.Sc. student at KUET, graduating December 2027.

WORK EXPERIENCE

Caretron — Embedded System Engineer | Dhaka, Bangladesh (Hybrid) *Aug 2025 – Present*

- Member of the team that developed Kairon, a digital medication reminder released in Digital and Smart IoT variants.
- Contributed to R&D, design and inspection, working in Altium Designer and SolidWorks.

FabLab KUET — Student Operator | Khulna, Bangladesh *Jul 2026 – Present*

- Run day-to-day lab operations, maintain equipment and support student projects.
- Lead hands-on workshops and training sessions on PCB design and fabrication.

EasyEDA — Official Campus Ambassador | KUET, Khulna *Jul 2026 – Present*

- Represent EasyEDA at KUET; course instructor for the official EasyEDA PCB training camp (Oct 2026).

Transform AI — PCB Designer (Project-based Contract) | Khulna, Bangladesh *May 2026 – Aug 2026*

- Designed the EDGE NODE, a four-board stacked assembly (main controller, power, relay interface and SBC carrier), with outlines, connectors and mounting verified together in 3D.
- Designed isolated Modbus, isolated RS232 and user-access boards; released the RS232 board for fabrication with Gerber, BOM and pick-and-place outputs.
- Wrote a power study comparing regulator ICs and direct battery powering for an IP camera system.

AuraXIoT — Junior Embedded System Engineer | Dhaka, Bangladesh *Apr 2025 – Jan 2026*

- Core R&D contributor to an Automatic Transfer Switch (ATS) product with 2,000+ units deployed in the field.
- Contributed to end-to-end hardware development: schematic design, PCB layout, enclosure development and BOM preparation.
- Designed for local manufacturing constraints in Bangladesh: PCB fabrication, assembly tolerances and injection-molded enclosures.
- Supported production testing and pre-delivery validation to ensure unit-level reliability.

FabLab KUET — Equipment Restoration Contributor | Khulna, Bangladesh *Oct 2024 – Feb 2025*

- Restored a non-functional Cirqoid CNC PCB mill by diagnosing faults, modifying the bed and building a custom G-code workflow around its software limits.
- Achieved consistent 0.15 mm trace and clearance, restoring in-house PCB prototyping for the lab.

Firefly High Tech — Project Trainee | Khulna, Bangladesh *Aug 2023 – Oct 2025*

- Helped develop and debug 30+ embedded and electronics projects for KUET students, covering sensors, microcontrollers and power systems.

Dept. of Energy Science and Engineering, KUET — Volunteer Research Contributor | Khulna, Bangladesh *Jan 2024 – Oct 2024*

- Built the experimental setup for geothermal research, installing thermocouples in underground test columns, and collected data across multiple configurations.

PRODUCTS AND PROJECTS

Commercial products

Kairon, Digital Medication Reminder (Caretron) | Altium Designer, SolidWorks *2026*

Healthcare product by Caretron: a three-drawer medication reminder for morning, noon and night doses, released in Digital and Smart IoT variants. Kairon is a team product; I was a member of the development team, contributing to R&D, design and inspection.
Portfolio: simul-ahmed.vercel.app/projects/kairon

ATS Power Control System (AuraXIoT) | Altium Designer, Proteus, SolidWorks *Jan 2026 – Apr 2026*

Commercial power-control unit for automatic mains-to-IPS switching in small backup systems (up to ~200 W). Solved instability during grid restoration and protected the IPS charging system. Designed the full schematic and double-sided production PCB; 2,000+ units manufactured.
Portfolio: simul-ahmed.vercel.app/projects/ats-power-control

Projects

Line-Following Robot: System Identification and PID | STM32F411, MATLAB, Altium Designer *Sep 2024 – Apr 2026*

Four generations of one robot, from a milled STM32F411 board to a research platform. Identified a second-order discrete-time plant model from Bluetooth telemetry (56.1% fit) and tuned the PID in MATLAB; under 5% overshoot on hardware.
Portfolio: simul-ahmed.vercel.app/projects/line-following-robot

Cirqoid CNC Controller Redesign | GRBL, stepper drivers *Jun 2026 – Present*
Designing a replacement GRBL controller for a PCB mill whose manufacturer no longer exists: four stepper drivers, limit switches and isolation, verified against the original hardware. Reverse-engineered the BLDC spindle driver; first prototype built.
Portfolio: simul-ahmed.vercel.app/projects/cirqoid-cnc

IoT Water Temperature Controller | ESP32, Altium Designer *Jan 2025 – Feb 2025*
ESP32 temperature monitor and relay heater control with on-board 12 V to 5 V regulation, taken from a module-based prototype to an integrated double-sided board.
Portfolio: simul-ahmed.vercel.app/projects/iot-water-temperature-controller

BCD Digital Clock | Altium Designer, Proteus *Jun 2025 – Jul 2025*
Fully discrete HH:MM:SS clock with no microcontroller, built from 74HC BCD counters and logic gates; verified in Proteus and laid out in Altium.
Portfolio: simul-ahmed.vercel.app/projects/bcd-digital-clock

5×5×5 LED Cube Panel and Driver Board | Altium Designer, Arduino Nano *Apr 2025 – May 2025*
Modular LED cube with layered 5×5 matrices and a shift-register driver board for synchronised pattern rendering.
Portfolio: simul-ahmed.vercel.app/projects/led-cube

EDUCATION

Khulna University of Engineering & Technology (KUET) *Jan 2023 – Present*
B.Sc. Engineering, Energy Science and Engineering. Expected graduation: December 2027.
Thesis (ongoing): experimental comparison of P&O and Incremental Conductance MPPT on a custom PV battery charger.

Police Lines School and College, Kushtia *Jan 2019 – Jan 2022*
HSC, Science

CERTIFICATIONS

Altium Skilled Engineer (Credly badge), Altium *Oct 2026*
Altium Workspace User Management, Altium (ID: afj4crzof5) *Oct 2026*
Basic Embedded Systems Design, Altium (ID: zswco0vxuy) *Sep 2026*
Altium Designer Essentials, On Demand, Altium (ID: bgxkfw5e1p) *Jul 2026*
Altium Emerging Engineer (Credly badge), Altium *Jul 2026*

TRAINING

SARBS Communication Ltd., Dhaka | Industrial Attachment: Product-level PCB Design in Altium Designer *Feb 2025 – Mar 2025*
- Analysed commercial schematics and built a complete thermocouple data logger schematic from modular open-source hardware blocks.

SARBS Communication Ltd., Dhaka | Industrial Attachment: PCB Fabrication (CNC Milling and Chemical Etching) *Oct 2024*
- Designed, milled and assembled a temperature-controlled fan controller, and etched a compact 3.3 V regulator board.

Orion Power Rupsha Ltd., Khulna | Industrial Attachment: Power Plant Operations *Nov 2025 – Dec 2025*
- Observed power generation, electrical distribution and operational safety procedures with the Operation Department.

ACHIEVEMENTS

2nd Place, Mars Autonomous Rover Rally Design Competition *Dec 2025*
Team Durbar (KUET), 2nd internationally. Organised by the NASA Texas Space Grant Consortium and Texas A&M University.

Top 10 Finalist, VLSITHON 3.0 *Sep 2026*
National chip-design competition by Ulkasemi; grand finalist with team Magical Musketeer.

Bangladesh Physics Olympiad, Regional Round (Category C) *2018, 2019*
7th position (2019) and 9th position (2018), Kushtia region; national round participant both years.

SKILLS

PCB Design	Altium Designer (certified), KiCad, EasyEDA: schematic capture, double-sided layout, multi-board assemblies, isolated and mixed-signal layout, fabrication outputs
Embedded	STM32, ESP32, ATmega (AVR), Arduino; firmware-aware layout decisions; MQTT, Bluetooth
Fabrication & CAM	CNC PCB milling, chemical etching, 3D printing, laser engraving; FlatCAM, Candle, bCNC, custom G-code
Simulation & Control	Proteus, MATLAB / Simulink, system identification, PID design
Mechanical	SolidWorks (enclosure design)
Programming	C / C++, Python, MATLAB
Sourcing	LCSC / JLCPCB and the Bangladeshi local market; cost-optimised, manufacture-ready BOMs
Languages	Bangla (native), English

REFERENCES

Dr. Md. Hasan Ali, Professor, Dept. of Energy Science and Engineering, KUET
Contact details available on request.

Omar Faruk Badhon, Embedded Systems Engineer, SARBS